FP Phase 3: Select the optimal algorithm, fine-tune and submit a final report

Start Assignment

- Due Apr 18 by 11:59pm
- Points 100
- Submitting a text entry box, a website url, a media recording, or a file upload
- Available Mar 4 at 12:30am - Apr 20 at 8:59pm

FP Phase 3 - experiment, fine-tune, select the optimal pipeline, and submit a final report

***Note:** Focus on moving fast (Speed matters in business) and on moving the needle (impact matters): focus on adding features/models/pipelines that bring value. If possible, try to do experiments on a subset of the data to get an indication of improvement before committing to full-scale experimentation.*

After your Mid-Evaluation, your client ask for a deep learning model! Check the objectives to understand what to do!

Note: By design, every aspect of the final project is up to the interpretation of your team. Don't think of the objectives as a checklist! The only things required for Phase 03 are the MLP implementation, the time-series' Features, the Graph's Features, and a leakage discussion.

Objectives

In Phase 3, please explore some or all of the following and report your findings:

1. Consider more sophisticated models: Random Forests, Gradient Boosted Decision Trees, multilayer perceptron (MLP) neural networks or other types of models - These are all suggestions, you have **free reign!**:
    1. Consider early stopping under different conditions (using a validation set) for both GBDT/NN.
        1. What data do you use (Hint: not training and not the held-out test set) for early stopping? What metrics do you use? ML metrics? Business metrics?
        2. Do you quit at the first sign of poor performance, or do you persevere for a few more epochs/trees?
        3. How does early stopping help?
    2. Train a MLP Neural Network - **This is the ask of the client! You have to comply!**
       -- Implement a multilayer perceptron (MLP) Neural Network (NN) model
       -- Experiment with at least 2 different MLP Network architectures (e.g., say, one with one hidden layer and a second with two hidden layers).
    3. 1. The following is a sample multilayer perceptron (MLP) model in PySpark:
        1.
```
# specify layers for the neural network:
# input layer of size 4 (features), two intermediate of size 5 and 4
# and output of size 2 (classes)
layers = [4, 5, 4, 2]
```

```
# create the trainer and set its parameters
trainer = MultilayerPerceptronClassifier(maxIter=100, layers=layers, blockSize=128, seed=1234)

# train the model
model = trainer.fit(train)

# compute accuracy on the test set
result = model.transform(test)
```

1. For more background and a template pipeline, please see here: [https://spark.apache.org/docs/latest/ml-classification-regression.html#multilayer-perceptron-classifier (https://spark.apache.org/docs/latest/ml-classification-regression.html#multilayer-perceptron-classifier)](https://spark.apache.org/docs/latest/ml-classification-regression.html#multilayer-perceptron-classifier)

-- **Extra Credit (Up to 5 points) if you are able to run other NN or Deep Learning technique - This NOT the ask of the client, so is Extra Credit.**

2. Fine-tune your pipeline using a grid search, or any search techniques (e.g., machine learning hyperparameters, pipeline steps)
   1. Is there a difference in performance? Is it related to features? Is it related to noise? What is impacting the model performance?
3. Do experiments on ALL the data for new features and experimental settings
   1. Use 2019 data as a blind test set that is never consulted during training.
   2. Report evaluation metrics in terms of cross-fold validation over the training set (2015-2018)
   3. Report evaluation metrics in terms of the 2019 dataset
   4. Report cluster size and experiment times for training and blind set evaluation
4. Feature engineering and refinement - **The client is very bullish in time series and graph theory, so you HAVE to include at least one Time-based feature and at least one Graph feature.**
   1. Time-based feature, e.g., recency, frequency, monetary (RFM)
   2. Graph-based features (e.g., a topic-based PageRank by airline)
   3. Event-based features (holidays, SuperBowl)
   4. Airport maintenance
   5. Natural disasters
   6. Airline Reputation
5. Exciting novel directions that you pursued. Some OPTIONS (none of this is mandatory, just to give you extra ideas)
   1. Data for all years are available here (new format and data dictionary)
      1. [https://www.ncei.noaa.gov/products/land-based-station/local-climatological-data (https://www.ncei.noaa.gov/products/land-based-station/local-climatological-data)](https://www.ncei.noaa.gov/products/land-based-station/local-climatological-data)
   2. Did you pull the data for recent data? (**EXTRA CREDIT - Up to 10 points of Extra Credit if you run predictions on 2020 and 2021 and do data/model drift analysis)**
   3. Multitask loss function (e.g., delayed versus not delayed AND the number of minutes delayed)
6. Clean up your code
   1. Try to have an end-to-end pipeline (HINT: Will your pipeline work with unseen data? it should!)

Final Report Requirements

- For this phase, please turn your business write-up in a notebook report.
  - This final report should be one complete, well-manicured notebook that tells a story. It should have a beginning, middle, and end.
  - It is expected that you have done many experiments. What would like to see is a summary of the key experiments (successes and surprises)
  - In short, your report should be business like in structure and language. Please try to use tables and diagrams as much as possible to shed light on and support your findings.
  - In your final report, write a **gap analysis** of your best pipeline.
  - Finally, the abstract, the discussion, and the conclusions are key. So please devote time to fleshing these out carefully.
  - NO CODE IN THIS NOTEBOOK!

DELIVERABLES:

1. **Your project notebook for this phase.**
2. **In-class Presentation**

Submission Requirements and format

You will submit the following items:

1. DataBricks notebook in .html format
   - Please make sure the links/images are visible to others by viewing your file in incognito mode on your browser
   - Your submission will be penalized if you do NOT meet these requirements

Final Project: Phase 4 (1)

Criteria	Ratings			Pts
In-Class Presentation In-Class Presentation should have a logical and business flow to it. In more detail, your In-Class Presentation should have a logical and scientific flow to it with main sections for each of the following: -- a title slide (with the project name, Group Number, the team member names, and photos). -- an abstract slide -- Make sure it has an outline slide with good descriptive section headings -- Team names, photos -- Project description -- Some summary visual EDA -- Feature engineering and Top features -- Overview of Modeling Pipelines explored -- Results and discussion of results (Accuracy, ROC/AUC, etc.. from this phase and previous phases) -- Conclusions (best performing model, number of features, top 10 best features, hyper-parameters) and next steps	**10 pts** **Full Marks!** **Great video!**	**5 pts** **Flow of In-Class Presentation not perfect** Your In-Class Presentation is hard to follow or lacks important details. Please make sure your presentation has a title slide (with the project name, Group Number, the team member names, and photos) has outline slide with good descriptive section headings. -- your presentation has a logical scientific manner: You need to cover the following in your presentation: -- Team member names,Group Number, photos -- Project description -- Some summary visual EDA based on Phase 2 findings -- Overview of Modeling Pipelines explored -- Results and discussion of results (Accuracy, ROC/AUC, etc.) -- Conclusion and next steps	**0 pts** **No Marks: In-Class Presentation does not meet my expectations for young data scientist.** Your video does not meet my expectations for young a data scientist team. Please review what was expected as a submission for this phase of the project. And reconcile where you fell short.	10 pts

Criteria	Ratings			Pts
Team and project meta information. Please provide the following: * Team ID * The complete list of team members and project meta information (e.g., email). * Credit assignment plan updates (who does/did what and when, amount of effort in terms of person-hours, start and end dates, estimated improvement in terms of key metrics) in Table format No Credit assignment plan means ZERO points A credit assignment plan not in Table format means ZERO points No start and end dates and (budgeted) hours of effort mean an incomplete plan. This may result in zero points.	**10 pts** **Complete team and project information.**	**5 pts** **Team and project meta information, or credit assignment table is incomplete or not formatted correctly** Team and project meta information is incomplete and or not formatted correctly To improve the quality of this section try the following: Please make a section and subsection for each of the following. - Team information -- Project Title -- Group number -- Names, emails -- Team member photos -- format the photos to be the same size and label each photo with the persons name -- Or merge all photos to make a montage The credit assignment table is incomplete or not formatted correctly	**0 pts** **No Marks: this does not meet my expectations for young data scientist.** This section does not meet my expectations for young a data scientist. Communication is key in data science and not getting this section right is not acceptable. Please review what was expected as a submission for this phase of the project. And reconcile where you fell short.	10 pts
Project Abstract Final Abstract: The final form of the abstract! It should have everything	**10 to >5.0 pts** **Full Marks! Excellent abstract!**	**5 to >0.0 pts** **Abstract is incomplete and or is not**	**0 pts** **No Marks: your abstract does not meet my**	10 pts

Criteria	Ratings			Pts
covered in previous phases, plus the new experiments and the final model selected, as well as your final results (report the number!)		**formatted correctly** Abstract is incomplete and or is not formatted correctly To improve the quality of this section. Then make sure to describe what your focused on and accomplished in this project (include this phase and previous phases). Have a look at the expectations with regard to a good abstract.	**expectations for young data scientist.** Your does not meet my expectations for young a data scientist. Communication is key in data science and not providing a good abstract is not acceptable. Please review what is expected as an abstract for this phase of the project and reconcile where you fell short.	
Data and feature engineering Please provide the following here: -- Summarize the data lineage and key data transformations (joins) -- List of feature families explored and explanation of each -- List of features within each family and description of each, along with THEIR EDA -- Please refer to experiments showing the value of each feature/family	**10 pts** **Full Marks! Excellent data lineage, new features, and EDA of new features** Full Marks! Excellent data lineage, new features, and EDA of new features!!	**5 pts** **Data Lineage or EDA of new features is incomplete and/or is not formatted correctly** Data Lineage is incomplete and as a result the experiments are not reproducable. EDA of new features is incomplete and/or is not formatted correctly for the following: -- List of feature families explored and explanation of each -- List of features within each family and description of	**0 pts** **No Marks: Data lineage and new features EDA does not meet my expectations for a young data scientist.** Data lineage is not provided. Your new features EDA does not meet my expectations for a young data scientist. Communication is key in data science and not providing a good new features EDA is not acceptable. Please review what is expected as a project description for this phase of the	10 pts

Criteria	Ratings				Pts
			each, along with their EDA -- Experiments showing the value of each feature/family	project and reconcile where you fell short.	
Neural Network (MLP) You are expected to train a Neural Network -- Implement Neural Network (NN) model -- Experiment with at least 2 different Network architectures and report results. For more background please see here: [https://spark.apache.org/docs/latest/ml-classification-regression.html#multilayer-perceptron-classifier (https://spark.apache.org/docs/latest/ml-classification-regression.html#multilayer-perceptron-classifier)](https://spark.apache.org/docs/latest/ml-classification-regression.html#multilayer-perceptron-classifier)	**10 pts** **Excellent NN Model experiments**		**5 pts** **NN is incomplete and or is not formatted correctly** Some or all of the following does not meet expectations: --- NN experiment is incomplete (training not shown; perfromance scores not provided) --- Experimental training results are not presented correctly (training curves by epoch not presented)	**0 pts** **No Marks: your NN does not meet my expectations for young data scientist.** Your NN does not meet my expectations for a young data scientist. Please review what is expected as a NN for this phase of the project and reconcile where you fell short.	10 pts
Leakage -- Define what is leakage and provide a a hypothetical example of leakage -- Go through your Pipeline and check if there is any leakage. -- Are you violating any cardinal sins of ML? -- Describe how your pipeline does not suffer from any leakage problem and does not violate any cardinal sins of ML	**10 pts** **Great presentation of leakage.**	**5 pts** **Leakage presentation is incomplete** -- Your presentation of leakage is incomplete -- Your presentation of leakage is incorrect	**3 pts** **Your Pipeline has leakage problems** - Your Pipeline has leakage problems (e.g., computing PageRank on the entire flights dataset	**0 pts** **No Marks: Does not meet my expectations for young data scientist.** Your model has major leakage problems.	10 pts

Criteria	Ratings				Pts
			versus over historical data that is available prior to the flight time minus, say, two hours)		
Modeling Pipelines Expectations here are to provide the following in sections and subsections: Modeling Pipelines -- A visualization of the modeling pipeline (s) and subpipelines if necessary -- Families of input features and count per family -- Number of input features -- Hyperparameters and settings considered -- Loss function used (data loss and regularization parts) in latex -- Number of experiments conducted -- Experiment table with the following details per experiment: ----- Baseline experiment ----- Any additional experiments ----- Final model tuned ----- best results (1 to three) for all experiments you conducted with the following details ---- Computational configuration used ----- Wall time for each experiment	**10 pts** **Great pipeline** Great pipeline code, comments and description for the new features (each with their own sub section):	**5 pts** **incomplete and or running improperly** Modeling pipelines are incomplete and or running improperly To improve the quality of this section: Make sure to provide the following in sections and subsections: -- Provide a block diagram of your modeling pipeline -- Make sure your pipelines are documented -- Verify your pipelines are complete and execute properly -- Your experiments are not documented properly (the are missing accurate descriptions, performance metrics, computational	**0 pts** **No Marks: Does not meet my expectations for young data scientist.** Your Modeling Pipelines section does not meet my expectations for a young data scientist. Experimentation and implementation are core skills in being a data scientists. In your submission both of these skills are lacking. Please review what is expected as a Modeling Pipelines for this phase of the project and reconcile where you fell short.		10 pts

Criteria	Ratings			Pts
		times and cluster details)		
Results and discussion of results Expectations here are to provide the following: The goal of Discussion' section is present an interpretation of key results , which means explain, analyse, and compare them (results from all the phases). Often, this part is the most important, simply because it lets the researcher take a step back and give a broader look at all experiments conducted. Do not discuss any outcomes not presented in the results part.	**20 pts** **Great results presentation and discussion of results.**	**10 pts** **Results and discussion are incomplete and or not adequately formatted** Results and discussion are incomplete and or not adequately formatted To improve the quality of this section: Make sure to provide the following in sections and subsections: -- Your experiments are not properly enumerated/tabulated and discussed (they are missing accurate descriptions, performance metrics) -- Do not discuss results not substantiated in your experimental section above in the modeling pipelines -- No Gap Analysis was provided	**0 pts** **No Marks: your results and discussion does not meet my expectations for young data scientist.** Your results and discussion does not meet my expectations for a young data scientist. Experimentation presentation and discussion are core skills in being a data scientists. In your submission both of these skills are lacking. Please review what is expected as a results and discussion for this phase of the project and reconcile where you fell short.	20 pts
Conclusion Expectations here are to address the following following in your conclusion in a main section by itself: -- Restate your project focus and explain why it's important. Make sure that this part of the conclusion is concise and clear.	**10 pts** **Full Marks! Excellent Conclusion!**	**5 pts** **Conclusion are incomplete and or does not flow well.** Conclusion is incomplete and or does not flow well. To improve the quality of this section: -- Make	**0 pts** **No Marks: your conclusion does not meet my expectations for young data scientist.** Your conclusion does not meet my expectations for a young data	10 pts

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-- Restate your hypothesis (e.g., ML pipelines with custom features can accurately predict .....) -- Summarize main points of your project: Remind your readers your key points. (e.g, best features, best model, hyper-parameters and so on) -- Discuss the significance of your results -- Discuss the future of your project.		sure your conclusion is about 150 long or less. -- You did not address all the expected points in a conclusion adequately.	scientist. Communicating a conclusion is a core skill for a data scientist. Please review what is expected as content, style, and format for a conclusion for this phase of the project and reconcile where you fell short.	
Neural Network [EXTRA CREDIT] Train any other NN or Deep Learning using a CPU clusters in a distributed fashion.	**5 pts** **Excellent NN Model experiments**	**2.5 pts** **NN is incomplete and or is not formatted correctly** Some or all of the following does not meet expectations: --- NN experiment is incomplete (training not shown; perfromance scores not provided) --- Experimental training results are not presented correctly (training curves by epoch not presented)	**0 pts** **No Marks: your NN does not meet my expectations for young data scientist.** Your NN does not meet my expectations for a young data scientist. Please review what is expected as a NN for this phase of the project and reconcile where you fell short.	5 pts
Pull the data for recent years [EXTRA CREDIT] Provide a clean dataset for Flights and/or Weather for Years 2020-2023	**10 pts** **Full Marks!**		**0 pts** **No Marks:**	10 pts
				Total Points: 115